NOTES ON TIME REVERSAL OF DIFFUSIONS

IOANNIS KARATZAS *

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Abstract

We review the theory of time reversal of diffusion processes as developed by Haussmann & Pardoux (1986) and Pardoux (1986), and present some illustrative examples.

1 Introduction

It is very well known and understood that the Markov property, expressed schematically as

$$\mathbb{P}(A \,|\, B \cap C) \,=\, \mathbb{P}(A \,|\, C)$$

where A stands for "future", B for "past", and C for "present", is invariant under time reversal. To wit if, as posited by the above equality, the future is conditionally independent of the past given the present, then *the past is also conditionally independent of the future, given the present,* i.e.,

$$\mathbb{P}(B \,|\, A \cap C) \,=\, \mathbb{P}(B \,|\, C)\,;$$

and vice-versa. In other words, a Markov process remains a Markov process under time reversal.

On the other hand, it is also well known that the strong Markov property is not necessarily preserved under time reversal (e.g., Rogers & Williams (1994), p. 330), and neither is the semimartingale property (e.g., Walsh (1982)). The reason for this failure is the same in both cases: after time-reversal, "we may know too much". Thus, the following questions arise rather naturally:

Given a diffusion process (in particular, a strong Markov process with continuous paths and a semimartingale) $X(\cdot) = \{X(t),\, 0 \le t \le 1\}$ *with certain specific drift and dispersion characteristics, under what conditions might the time-reversed process* $\widehat{X}(\cdot)$ *given by*

$$\widehat{X}(t) \,:=\, X(1-t)\,, \qquad 0 \le t \le 1 \tag{1.1}$$

* Department of Mathematics, Columbia University, New York, NY 10027. E-Mail: *ik1@columbia.edu.*
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also be a diffusion? if it happens to be, what are the characteristics of the time-reversed diffusion? and when can we expect these characteristics to be identical to those of the original process $X(\cdot)$, *that is, not to be able to tell the original diffusion from its time-reversed cousin as far as finite-dimensional distributions are concerned?*

Such questions go back at least to Schrödinger (1931, 1932) and Kolmogorov (1937); they were dealt with systematically by Nelson (1967) (see also Carlen (1984)) in the context of Nelson's dynamical theories for Brownian motion and diffusion. There is now a rather nice theory that answers these questions and provides, as a kind of "bonus", some rather unexpected results as well. It was developed by workers in the theory of filtering, interpolation of extrapolation, where such issues arise naturally – most notably Anderson (1982), Castanon (1982), Haussmann & Pardoux (1986) and Pardoux (1986). I present this theory here in the spirit of the wonderful expository paper by P.A. Meyer (1994). Very interesting related results in a non-Markovian context, but with dispersion structure given by the identity matrix, have been obtained by Föllmer (1985, 1986). Once the theory has been developed, it is illustrated on several examples.

2 The Setting

Let us place ourselves on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathbb{F} = \{\mathcal{F}(t)\}_{0 \le t \le 1}$ rich enough to support an $\mathbb{R}^d-$valued Brownian motion $W(\cdot) = (W_1(\cdot), \cdots, W_d(\cdot))'$ adapted to $\mathbb{F}$, as well as an independent $\mathcal{F}(0)-$measurable random vector $\boldsymbol{\xi} = (\xi_1, \cdots, \xi_n)' : \Omega \to \mathbb{R}^n$. In fact, we shall assume that $\mathbb{F}$ *is* the natural filtration generated by these two objects, in the sense that we shall take

$$\mathcal{F}(t) \,=\, \sigma\big(\boldsymbol{\xi}, W(s)\,;\; 0 \le s \le t\big)\,, \qquad 0 \le t \le 1$$

modulo $\mathbb{P}-$augmentation. Next, we assume that the system of stochastic equations

$$X_i(t) \,=\, \xi_i + \int_0^t b_i\big(s, X(s)\big)\, ds \,+\, \sum_{\nu=1}^{d} \int_0^t \mathrm{s}_{i\nu}\big(s, X(s)\big)\, \mathrm{d}W_\nu(s)\,, \qquad 0 \le t \le 1\,, \tag{2.1}$$

for $i = 1, \cdots, n$ admits a pathwise unique, strong solution. It is then well known that the resulting continuous process $X(\cdot) = (X_1(\cdot), \cdots, X_n(\cdot))'$ is $\mathbb{F}-$adapted (the strong solvability of the equation (2.1)), which implies that we have also

$$\mathcal{F}(t) \,=\, \sigma\big(X(s),\, W(s)\,;\; 0 \le s \le t\big) \,=\, \sigma\big(X(0),\, W(t) - W(\theta)\,;\; 0 \le \theta \le t\big) \tag{2.2}$$

modulo $\mathbb{P}-$augmentation, for $0 \le t \le 1$; as well as that $X(\cdot)$ has the strong Markov property, and is thus a diffusion process with drifts $b_i(\cdot\,,\cdot)$ and dispersions $\mathrm{s}_{i\nu}(\cdot\,,\cdot)$, $i = 1, \cdots, n$, $\nu = 1, \cdots, d$. We shall denote by

$$\mathrm{a}_{ij}(t,x) \,:=\, \sum_{\nu=1}^{d} \mathrm{s}_{i\nu}(x)\, \mathrm{s}_{j\nu}(t,x)\,, \qquad 1 \le i,j \le n$$

the $(i,j)^{th}$ entry of the covariance matrix $\mathrm{a}(t,x) := \mathrm{s}(t,x)\,\mathrm{s}'(t,x)$.

These characteristics are given mappings from $[0,1] \times \mathbb{R}^n$ into $\mathbb{R}$ with sufficient smoothness; in particular, such that the probability density function $\mathfrak{p}_t(\cdot) : \mathbb{R}^n \to (0,\infty)$ in

$$\mathbb{P}\big(X(t) \in A\big) \,=\, \int_A \mathfrak{p}_t(x)\,\mathrm{d}x\,, \qquad A \in \mathcal{B}(\mathbb{R}^n)$$

is smooth. Sufficient conditions on the drift $b_i(\cdot\,,\cdot)$ and dispersion $\mathrm{s}_{i\nu}(\cdot\,,\cdot)$ characteristics that lead to such smoothness, are provided by the Hörmander hypoellipticiy conditions; see for instance Bell (1995), Nualart (1995) for this result, as well as Rogers (1985) for a very simple argument in the one-dimensional case ($n = d = 1$). We refer to Friedman (1975), Rogers & Williams (1987) or Karatzas & Shreve (1991) for the basics of the theory of stochastic equations of the form (2.1).

The probability density function $\mathfrak{p}(t,\cdot) \equiv \mathfrak{p}_t(\cdot) : \mathbb{R}^n \to (0,\infty)$ solves the forward Kolmogorov (1931) equation (Friedman (1975), p. 149)

$$\frac{\partial}{\partial t}\,\mathfrak{p}_t(x) \,=\, \frac{1}{2}\,\sum_{i=1}^n\sum_{j=1}^n D^2_{ij}\big(\mathrm{a}_{ij}(t,x)\,\mathfrak{p}_t(x)\big) - \sum_{i=1}^n D_i\big(b_i(t,x)\,\mathfrak{p}_t(x)\big)\,, \qquad (t,x) \in (0,\infty)\times\mathbb{R}^n. \tag{2.3}$$

If the drift and dispersion characteristics do not depend on time, and an invariant probalility measure exists for the diffusion process of (2.1), the density function $\mathfrak{p}(\cdot)$ of this measure solves the stationary version of this forward Kolmogorov equation, to wit

$$\frac{1}{2}\,\sum_{i=1}^n\sum_{j=1}^n D^2_{ij}\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\big) \,=\, \sum_{i=1}^n D_i\big(b_i(x)\,\mathfrak{p}(x)\big)\,, \qquad x \in \mathbb{R}^n\,. \tag{2.4}$$

3 Time Reversal

Consider now the filtration $\big\{\widehat{\mathcal{F}}(1-t)\big\}_{0\le t\le 1}$ given by

$$\widehat{\mathcal{F}}(1-t) \,:=\, \sigma\big(X(s),\, W(s)-W(t)\,;\; t \le s \le 1\big)\,, \qquad 0 \le t \le 1\,. \tag{3.1}$$

It is not hard to see that this filtration is expressed equivalently as

$$\begin{aligned}\widehat{\mathcal{F}}(1-t) \,&=\, \sigma\big(X(t),\, W(s)-W(t)\,;\; t \le s \le 1\big) \,=\, \sigma\big(X(t),\, W(s)-W(1)\,;\; t \le s \le 1\big)\\ &=\, \sigma\big(X(1),\, W(s)-W(t)\,;\; t \le s \le 1\big) \,=\, \sigma(X(1)) \vee \widehat{\mathcal{G}}(1-t)\,.\end{aligned} \tag{3.2}$$

Here

$$\widehat{\mathcal{G}}(1-t) \,:=\, \sigma\big(W(s)-W(t)\,;\; t \le s \le 1\big)\,, \qquad 0 \le t \le 1 \tag{3.3}$$

is the sigma-algebra of Brownian increments after time t, and is independent of the random vector $X(t)$. In particular, $\widehat{\mathcal{F}}(1-t)$ is generated by the terminal value $X(1)$ and by the increments of $W(\cdot)$ on $[t,1]$.

3.1 The Backwards Filtration

The time-reversed processes $\widehat{X}(\cdot)$ as in (1.1), as well as

$$\widetilde{W}(t) := W(1-t) - W(1), \qquad 0 \le t \le 1 \tag{3.4}$$

are both adapted to the *Backwards Filtration* $\widehat{\mathbb{F}} := \big\{ \widehat{\mathcal{F}}(t) \big\}_{0 \le t \le 1}$, where

$$\widehat{\mathcal{F}}(t) = \sigma\big(X(1-\theta),\, W(1-\theta) - W(1-t)\,;\; 0 \le \theta \le t\big) = \sigma\big(\widehat{X}(\theta),\, \widetilde{W}(\theta) - \widetilde{W}(t)\,;\; 0 \le \theta \le t\big)$$

from (3.1). Note that, by complete analogy with (2.2), we have also

$$\widehat{\mathcal{F}}(t) = \sigma\big(X(1),\, W(1-\theta) - W(1-t)\,;\; 0 \le \theta \le t\big) = \sigma\big(\widehat{X}(0)\big) \vee \widehat{\mathcal{G}}(t)$$

on account of (3.2), where

$$\widehat{\mathcal{G}}(t) = \sigma\big(W(1-\theta) - W(1-t)\,;\; 0 \le \theta \le t\big) = \sigma\Big(\widetilde{W}(\theta) - \widetilde{W}(t)\,;\; 0 \le \theta \le t\Big). \tag{3.5}$$

In words: the sigma-algebra $\widehat{\mathcal{F}}(t)$ is generated by the terminal value $X(1)$ of the forward process (i.e., by the original value $\widehat{X}(0)$ of the backward process) and by the increments of the time-reversed process $\widetilde{W}(\cdot)$ on $[0,t]$; see the expressions right above. Furthermore, the sigma-algebra $\widehat{\mathcal{F}}(t)$ measures all the random variables $\widehat{X}(\theta)$, $\theta \in [0,t]$.

Remark 3.1. In fact, the process $\widetilde{W}(\cdot)$ *is a Brownian motion of the filtration* $\widehat{\mathbb{G}} := \{\widehat{\mathcal{G}}(t)\}_{0 \le t \le 1}$ as in (3.5), generated by the increments of $W(\cdot)$ after time $1-t$, $0 \le t \le 1$.

This is because it is a martingale with respect to this filtration, has continuous paths, and its quadratic variation is that of Brownian motion (the P. Lévy theorem). To see, for instance, the martingale property, observe that with $0 \le s \le t \le 1$ we have almost surely

$$\begin{aligned}\mathbb{E}\left[\,\widetilde{W}(1-s) - \widetilde{W}(1-t)\,\middle|\,\widehat{\mathcal{G}}(1-t)\,\right] &= \mathbb{E}\left[\,W(s) - W(t)\,\middle|\,W(\theta) - W(t),\ t \le \theta \le 1\,\right] \\ &= \mathbb{E}\big(W(s) - W(t)\big) = 0,\end{aligned}$$

thanks to the independence of the Brownian increments. This, however, does *not* imply

$$\mathbb{E}\left[\,\widetilde{W}(1-s) - \widetilde{W}(1-t)\,\middle|\,\widehat{\mathcal{F}}(1-t)\,\right] = \mathbb{E}\left[\,\widetilde{W}(1-s) - \widetilde{W}(1-t)\,\middle|\,\sigma\big(X(1)\big) \vee \widehat{\mathcal{G}}(1-t)\,\right] = 0.$$

In fact, in the next section we shall see that the process $\widetilde{W}(\cdot)$ *is only a semimartingale of the backwards filtration* $\widehat{\mathbb{F}}$, and identify its semimartingale decomposition.

4 Some Remarkable Brownian Motions

Following the exposition and ideas in Meyer (1994), we start with a couple of observations. First, for every $t \in [0,1]$ and every integrable, $\widehat{\mathcal{F}}(1-t)-$measurable random variable $\mathcal{K}$, we have

$$\mathbb{E}(\mathcal{K} \,|\, \mathcal{F}(t)) = \mathbb{E}(\mathcal{K} \,|\, X(t)), \quad \text{almost surely.} \tag{4.1}$$

Secondly, if we fix a function $G \in \mathcal{C}_0^\infty(\mathbb{R}^n)$ and a time-point $t \in (0,1]$, define

$$g(s,x) \,:=\, \mathbb{E}\left[\, G(X(t)) \,\middle|\, X(s)=x \,\right], \qquad (s,x) \in [0,t] \times \mathbb{R}^n \,,$$

and invoke the Markov property of $X(\cdot)$, we deduce that the process

$$g(s,X(s)) = \mathbb{E}\left[\, G(X(t)) \,|\, X(s) \,\right] = \mathbb{E}\left[\, G(X(t)) \,|\, \mathcal{F}(s) \,\right], \qquad 0 \le s \le t$$

is an $\mathbb{F}-$martingale and obtain

$$G(X(t)) - g(s,X(s)) = g(t,X(t)) - g(s,X(s)) = \sum_{i=1}^n \sum_{\nu=1}^d \int_s^t D_i g(u,X(u))\, \mathrm{s}_{i\nu}(u,X(u))\, \mathrm{d}W_\nu(u)\,.$$

For every index $\nu = 1, \cdots, d$ this gives, after integrating by parts,

$$\begin{aligned}
\mathbb{E}\left[\, \big(W_\nu(t) - W_\nu(s)\big) \cdot G(X(t)) \,\right] \,&=\, \mathbb{E}\left[\, \big(W_\nu(t) - W_\nu(s)\big) \cdot \big(g(t,X(t)) - g(s,X(s))\big) \,\right] \\
= \mathbb{E} \sum_{i=1}^n \int_s^t D_i g(u,X(u))\, \mathrm{s}_{i\nu}(u,X(u))\, \mathrm{d}u \,&=\, \sum_{i=1}^n \int_s^t \int_{\mathbb{R}^n} \big(D_i g \cdot \mathrm{s}_{i\nu}\big)(u,x)\, \mathfrak{p}_u(x)\, \mathrm{d}x\, \mathrm{d}u \\
&= \, -\sum_{i=1}^n \int_s^t \int_{\mathbb{R}^n} g(u,x)\, D_i\big(\mathfrak{p}_u(x)\, \mathrm{s}_{i\nu}(u,x)\big)\, \mathrm{d}x\, \mathrm{d}u \\
&= \, -\int_s^t \int_{\mathbb{R}^n} g(u,x)\, \frac{\mathrm{div}\big(\mathfrak{p}_u(x)\, \overline{\mathrm{s}}_\nu(u,x)\big)}{\mathfrak{p}_u(x)}\, \mathfrak{p}_u(x)\, \mathrm{d}x\, \mathrm{d}u \\
&= \, -\int_s^t \mathbb{E}\left[g\big(u,X(u)\big) \cdot \frac{\mathrm{div}\big(\mathfrak{p}\, \overline{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\, \big(u,X(u)\big) \right] \mathrm{d}u \\
&= \, -\mathbb{E}\left[G\big(X(t)\big) \cdot \int_s^t \frac{\mathrm{div}\big(\mathfrak{p}\, \overline{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\, \big(u,X(u)\big)\, \mathrm{d}u \right].
\end{aligned}$$

Here $\overline{\mathrm{s}}_\nu(u,\cdot)$ is the ν^{th} column vector of the dispersion matrix, and we have set

$$\mathrm{div}\big(\mathfrak{p}_u(x)\, \overline{\mathrm{s}}_\nu(u,x)\big) \,:=\, \sum_{i=1}^n D_i\big(\mathfrak{p}_u(x)\, \mathrm{s}_{i\nu}(u,x)\big)\,, \qquad \nu = 1, \cdots, d\,.$$

Comparing the first and last expressions in the above string of equalities, we see that with $0 \le s \le t$ we have

$$\mathbb{E}\left[G\big(X(t)\big) \cdot \left\{ W_\nu(t) - W_\nu(s) + \int_s^t \frac{\mathrm{div}\big(\mathfrak{p}\, \overline{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\, \big(u,X(u)\big)\, \mathrm{d}u \right\} \right] \,=\, 0 \tag{4.2}$$

for every $G \in \mathcal{C}_0^\infty(\mathbb{R}^n)$, and thus by extension for every bounded, measurable $G : \mathbb{R}^n \to \mathbb{R}$.

Theorem 4.1. *The vector process* $B(\cdot) = \big(B_1(\cdot), \cdots, B_d(\cdot)\big)'$ *defined as*

$$B_\nu(t) \,:=\, \widetilde{W}_\nu(t) - \int_0^t \frac{\operatorname{div}\big(\mathfrak{p}\,\bar{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\big(1-u, \widehat{X}(u)\big)\,\mathrm{d}u \tag{4.3}$$

$$= W_\nu(1-t) - W_\nu(1) - \int_{1-t}^1 \frac{\operatorname{div}\big(\mathfrak{p}\,\bar{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\big(v, X(v)\big)\,\mathrm{d}v\,, \qquad 0 \le t \le 1$$

for $\nu = 1, \cdots, d$, *is Brownian motion with respect to the backwards filtration* $\widehat{\mathbb{F}} = \big\{\widehat{\mathcal{F}}(t)\big\}_{0\le t\le 1}$.

Remark 4.1. The Brownian motion process $B(\cdot)$ is thus independent of $\widehat{\mathcal{F}}(0)$, and therefore also of the $\widehat{\mathcal{F}}(0)-$measurable random variable $X(1)$. A bit more generally,

$$\big\{B(1-s) - B(1-t),\ 0 \le s \le t\big\} \text{ is independent of } \ \widehat{\mathcal{F}}(1-t) \supseteq \sigma\big(X(u),\ t \le u \le 1\big). \tag{4.4}$$

Note also from (4.3) that

$$B_\nu(1-s) - B_\nu(1-t) \,=\, W_\nu(s) - W_\nu(t) - \int_s^t \frac{\operatorname{div}\big(\mathfrak{p}\,\bar{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\big(v, X(v)\big)\,\mathrm{d}v\,, \qquad 0 \le s \le t\,. \qquad \square$$

Reversing time once again, we obtain the following corollary of Theorem 4.1.

Corollary. *The* $\mathbb{F}$*-adapted vector process* $V(\cdot) = \big(V_1(\cdot), \cdots, V_d(\cdot)\big)'$ *with components*

$$V_\nu(t) \,:=\, B_\nu(1-t) - B_\nu(1) \,=\, W_\nu(t) + \int_0^t \frac{\operatorname{div}\big(\mathfrak{p}\,\bar{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\big(u, X(u)\big)\,\mathrm{d}u\,, \qquad 0 \le t \le 1 \tag{4.5}$$

for $\nu = 1, \cdots, d$, *is yet another Brownian motion (with respect to its own filtration* $\mathbb{F}^V \subset \mathbb{F}$ *).*

This process is independent of the random variable $X(1)$*; and a bit more generally, for every* $t \in (0, 1]$, *the* $\sigma-$*algebra*

$$\mathcal{F}^V(t) := \sigma(V(\theta)\,;\ 0 \le \theta \le t) \tag{4.6}$$

generated by present-and-past values of $V(\cdot)$, *is independent of* $\sigma(X(u),\ t \le u \le 1)$, *the* $\sigma-$*algebra generated by present-and-future values of* $X(\cdot)$.

Proof of Theorem 4.1: It suffices to show that each $B_\nu(\cdot)$ is a martingale of $\widehat{\mathbb{F}}$; because then, in view of the continuity of paths and the easily checked property $\langle B_\nu, B_\ell\rangle(t) = t\,\delta_{j\ell}$, we can deduce that each $B_\nu(\cdot)$ is a Brownian motion in the backwards filtration $\widehat{\mathbb{F}}$ (and of course also in its own filtration), and that $B_\nu(\cdot)$, $B_\ell(\cdot)$ are independent for $\ell \neq j$, appealing to the P. Lévy theorem once again. Now for $0 \le s \le t \le 1$ we have to show

$$\mathbb{E}\left[\,\big\{B_\nu(1-s) - B_\nu(1-t)\big\}\cdot\mathcal{K}\,\right] \,=\, 0$$

for every bounded, $\widehat{\mathcal{F}}(1-t)-$measurable random variable $\mathcal{K}$, or equivalently

$$\mathbb{E}\left[\mathbb{E}\big(\mathcal{K}\,|\,\mathcal{F}(t)\big)\cdot\left\{W_\nu(t) - W_\nu(s) + \int_s^t \frac{\operatorname{div}\big(\mathfrak{p}\,\bar{\mathrm{s}}_\nu\big)}{\mathfrak{p}}\big(u, X(u)\big)\,\mathrm{d}u\right\}\right] \,=\, 0\,,$$

as the expression inside the braces is $\mathcal{F}(t)-$measurable. But recalling (4.1) we have

$$\mathbb{E}\big(\mathcal{K}\,|\,\mathcal{F}(t)\big) \,=\, \mathbb{E}\big(\mathcal{K}\,|\,X(t)\big) \,=\, G(X(t))$$

for some bounded, measurable $G : \mathbb{R}^n \to \mathbb{R}$, and the desired result follows from (4.2). $\square$

4.1 The case of unit dispersion matrix, and the Jeulin-Yor process

Let us concentrate now on the case where the dispersion in (2.1) is the unit $(n \times n)$ matrix $\mathrm{s} \equiv I_n$, $d = n$, namely,

$$X(t) \,=\, \boldsymbol{\xi} + \int_0^t b\big(s, X(s)\big)\, \mathrm{d}s + W(t)\,, \qquad 0 \le t \le 1\,. \tag{4.7}$$

The process of (4.5) takes then the form

$$V(t) \,=\, X(t) - X(0) + \int_0^t \Big(D \log \mathfrak{p} - b\Big)\big(s, X(s)\big)\, \mathrm{d}s\,, \qquad 0 \le t \le 1\,. \tag{4.8}$$

It is clearly adapted to the filtration $\mathbb{F}^X$ generated by the process $X(\cdot)$ as in (4.6), namely

$$\mathbb{F}^V := \big(\mathcal{F}^V(t)\big)_{0 \le t \le 1} \subseteq \big(\mathcal{F}^X(t)\big)_{0 \le t \le 1} =: \mathbb{F}^X\,; \tag{4.9}$$

as indicated in the Corollary, each $\mathcal{F}^V(t)$ is independent of $\sigma(X(u),\ t \le u \le 1)$, and in particular of $X(t)$ and of $X(1)$. As a consequence of these considerations, the filtration inclusion $\mathcal{F}^V(t) \subseteq \mathcal{F}^X(t)$ from (4.9) is *strict:*

$$\mathcal{F}^V(t) \subsetneqq \mathcal{F}^X(t)\,, \qquad 0 \le t \le 1\,. \tag{4.10}$$

Thus, in addition to the $\mathbb{F}-$semimartingale representation in (4.7), we have in this case for $X(\cdot)$ also the representation

$$X(t) \,=\, \boldsymbol{\xi} - \int_0^t \Big(D \log \mathfrak{p} - b\Big)\big(s, X(s)\big)\, \mathrm{d}s + V(t)\,, \qquad 0 \le t \le 1 \tag{4.11}$$

on account of (4.8).

Now suppose that the initial condition $\boldsymbol{\xi} \equiv \xi \in \mathbb{R}^n$ is degenerate. Then the strong solvability of (2.1) implies $\mathbb{F}^X \equiv \mathbb{F}^W$: the filtration $\mathbb{F}^X$ generated by the process $X(\cdot)$ *is* generated by *some* Brownian motion, namely the driver $W(\cdot)$ of the stochastic equation (4.7).

On the other hand, and on account of (4.11), the process $V(\cdot)$ is adapted to the filtration $\mathbb{F}^X$;. It is also a Brownian motion with respect to *its own, strictly smaller, filtration* $\mathbb{F}^V$; and (4.10) shows that *the stochastic equation* (4.11) *does not admit a strong solution.*

Example 4.1. The Jeulin-Yor process: With $n = 1$, $\xi \equiv 0$, $\mathrm{s} \equiv 1$ and $b \equiv 0$, we have that $X(\cdot) = W(\cdot)$ is a standard, one-dimensional Brownian motion, and

$$\mathfrak{p}(t,x) \,\equiv\, \mathfrak{p}_t(x) \,=\, \big(2\pi t\big)^{-1/2} \exp\left\{ -\frac{x^2}{2\,t} \right\}.$$

We deduce from the Corollary to Theorem 4.1 that the process of (4.8), namely, *the Jeulin-Yor (1990) process*

$$\boxed{V(t) \,:=\, X(t) - \int_0^t \frac{X(s)}{s}\, \mathrm{d}s\,, \qquad 0 \le t \le 1\,,} \tag{4.12}$$

is Brownian motion, and is independent of the random variable $X(1)$. I still find this result rather remarkable, almost twenty five years since I became aware of it.

In fact, for each $t \in [0,1]$, the random variable $X(t) = W(t)$ is independent of the $\sigma-$algebra $\mathcal{F}^V(t) = \sigma(V(\theta),\, 0 \leq \theta \leq t)$. Once again, the analogue of (4.11), namely, *the stochastic equation*

$$X(\cdot) \,=\, \int_0^{\cdot} \frac{X(t)}{t}\,\mathrm{d}t \,+\, V(\cdot)\,,$$

cannot possibly have a strong solution.

With some justification, then, we shall call the process of (4.8) *a "generalized Jeulin-Yor (1990) process".*

Remark 4.2. Here is another example in the same spirit. Suppose the random variable $Z \sim \mathcal{N}(0,1)$ is independent of the Brownian motion $W(\cdot)$. Then, for every $k \in (0,\infty)$, and setting $X_k(t) := W(t) + k\,Z\,t$, the process

$$\begin{aligned} N_k(t) \,&:=\, X_k(t) - \int_0^t \frac{X_k(s)}{(1/k^2)+s}\,\mathrm{d}s \,=\, W(t) + k\,Z\,t - \int_0^t \frac{W(s)+k\,Z\,s}{(1/k^2)+s}\,\mathrm{d}s \\ &=\, W(t) + \frac{Z}{k}\,\log\big(1+k^2\,t\big) - \int_0^t \frac{W(s)}{(1/k^2)+s}\,\mathrm{d}s\,, \qquad 0 \leq t \leq 1 \end{aligned}$$

is again standard Brownian motion. Letting $k \uparrow \infty$ we recover in the limit the Jeulin-Yor process of (4.12).

This remark is inspired by Example 4.3.5, pp. 128-131 in Davis (1977). It has its origins in the context of filtering theory, where $N_k(\cdot)$ acquires the significance of the celebrated "innovations process". Example 6.4 and, in particular Remark 6.3 below, provide further elaboration on this remark, and offer details about the claims made here.

5 The Diffusion Property under Time Reversal

Let us return now to the question, whether the time-reversed process $\widehat{X}(\cdot)$ of (1.1), (2.1) is a diffusion. We start by expressing $X_i(\cdot)$ of (2.1) in terms of a Backward Itô integral (see section 9) as

$$\begin{aligned} X_i(t) - \xi_i - \int_0^t b_i\big(s, X(s)\big)\,\mathrm{d}s \,&=\, \sum_{\nu=1}^{d} \int_0^t \mathrm{s}_{i\nu}\big(s, X(s)\big)\,\mathrm{d}W_\nu(s) \\ &=\, \sum_{\nu=1}^{n} \bigg(\int_0^t \mathrm{s}_{i\nu}\big(s, X(s)\big) \bullet \mathrm{d}W_\nu(s) - \big\langle \mathrm{s}_{i\nu}(\cdot\,, X(\cdot)), W_\nu(\cdot)\big\rangle(t) \bigg)\,. \end{aligned}$$

From (2.1), we have by Itô's rule that the process

$$\mathrm{s}_{i\nu}(\,\cdot\,, X(\cdot)) - \mathrm{s}_{i\nu}(0\,, \boldsymbol{\xi}) - \sum_{j=1}^{n}\sum_{\nu=1}^{d} \int_0^{\cdot} D_j\, \mathrm{s}_{i\nu}(t\,, X(t)) \cdot \mathrm{s}_{j\nu}(t\,, X(t))\,\mathrm{d}W_\nu(t)$$

is of finite first variation, therefore

$$\big\langle \mathrm{s}_{i\nu}(\cdot\,,X(\cdot)), W_\nu(\cdot)\big\rangle(t) \;=\; \sum_{j=1}^{n}\int_0^t \mathrm{s}_{j\nu}(s\,,X(s))\, D_j\, \mathrm{s}_{i\nu}(s\,,X(s))\,\mathrm{d}s\,.$$

We conclude

$$X_i(t) \;=\; \xi_i - \int_0^t \Big(\sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}\, D_j\, \mathrm{s}_{i\nu} - b_i\Big)\big(s, X(s)\big)\mathrm{d}s + \sum_{\nu=1}^{d}\int_0^t \mathrm{s}_{i\nu}\big(s, X(s)\big) \bullet \mathrm{d}W_\nu(s)\,.$$

Evaluating also at $t=1$, then subtracting, we obtain

$$X_i(t) \;=\; X_i(1) + \int_t^1 \Big(\sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}\, D_j\, \mathrm{s}_{i\nu} - b_i\Big)\big(s, X(s)\big)\mathrm{d}s - \sum_{\nu=1}^{d}\int_t^1 \mathrm{s}_{i\nu}\big(s, X(s)\big) \bullet \mathrm{d}W_\nu(s)\,,$$

as well as

$$\widehat{X}_i(t) \;=\; \widehat{X}_i(0) + \int_0^t \Big(\sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}\, D_j\, \mathrm{s}_{i\nu} - b_i\Big)\big(1-s, \widehat{X}(s)\big)\mathrm{d}s + \sum_{\nu=1}^{d}\int_0^t \mathrm{s}_{i\nu}\big(1-s, \widehat{X}(s)\big)\, \mathrm{d}\widetilde{W}_\nu(s)$$

by reversing time. Note that the backward Itô integral for $W(\cdot)$ becomes a forward Itô integral for the process $\widetilde{W}(\cdot)$, the time-reversal of $W(\cdot)$ in the manner of (3.4).

But now let us recall (4.3), on the strength of which the above expression takes the form

$$\widehat{X}_i(t) \;=\; \widehat{X}_i(0) + \sum_{\nu=1}^{d}\int_0^t \mathrm{s}_{i\nu}\big(1-s, \widehat{X}(s)\big)\, \mathrm{d}B_\nu(s)$$
$$+ \int_0^t \Big(\sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}\, D_j\, \mathrm{s}_{i\nu} + \sum_{\nu=1}^{d} \mathrm{s}_{i\nu}\, \frac{\mathrm{div}\big(\mathfrak{p}\, \overline{\mathrm{s}}_\nu\big)}{\mathfrak{p}} - b_i\Big)\big(1-s, \widehat{X}(s)\big)\, \mathrm{d}s\,, \quad 0 \le t \le 1.$$

But in conjunction with Theorem 4.1, this means that the time-reversed process $\widehat{X}(\cdot)$ of (1.1), (2.1) is a semimartingale of the backwards filtration $\widehat{\mathbb{F}}$, with decomposition

$$\boxed{\widehat{X}_i(t) \;=\; \widehat{X}_i(0) + \int_0^t \widehat{b}_i\big(1-s, \widehat{X}(s)\big)\, \mathrm{d}s \;+\; \sum_{\nu=1}^{d}\int_0^t \mathrm{s}_{i\nu}\big(1-s, \widehat{X}(s)\big)\, \mathrm{d}B_\nu(s)} \tag{5.1}$$

for $0 \le t \le 1$ where, for each $i = 1, \cdots, n$, the function $\widehat{b}_i(\cdot\,,\cdot)$ is specified by

$$\widehat{b}_i(t,x) + b_i(t,x) \;=\; \sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}(t,x)\;D_j\, \mathrm{s}_{i\nu}(t,x) + \sum_{\nu=1}^{d} \mathrm{s}_{i\nu}(t,x)\, \frac{\mathrm{div}\big(\mathfrak{p}_t(x)\, \overline{\mathrm{s}}_\nu(t,x)\big)}{\mathfrak{p}_t(x)}$$

$$= \sum_{j=1}^{n}\sum_{\nu=1}^{d} \mathrm{s}_{j\nu}(t,x)\;D_j\, \mathrm{s}_{i\nu}(t,x) + \sum_{\nu=1}^{d} \frac{\mathrm{s}_{i\nu}(t,x)}{\mathfrak{p}_t(x)}\Big(\sum_{j=1}^{n} D_\ell\big(\mathfrak{p}_t(x)\, \mathrm{s}_{j\nu}(t,x)\big)\Big)$$

$$= \sum_{j=1}^{n} \sum_{\nu=1}^{d} \mathrm{s}_{j\nu}(t,x)\; D_j\, \mathrm{s}_{i\nu}(t,x) + \mathrm{s}_{i\nu}(t,x) \Big(D_j\, \mathrm{s}_{j\nu}(t,x) + \mathrm{s}_{j\nu}(t,x)\, D_j \log \mathfrak{p}_t(x) \Big) \Big\}$$

$$= \sum_{j=1}^{n} \Big(D_j\, \mathrm{a}_{ij}(t,x) + \mathrm{a}_{ij}(t,x) \cdot D_j \log \mathfrak{p}_t(x) \Big) \,.$$

Theorem 5.1. Generalized Nelson Equation: *Under the assumptions of this note, the time-reversed process* $\widehat{X}(\cdot)$ *of* (1.1), (2.1) *is a diffusion in the backwards filtration* $\widehat{\mathbb{F}}$ *, with characteristics as in* (5.1)*, namely, dispersions* $\mathrm{s}_{i\nu}(1-t,x)$ *and drifts* $\widehat{b}_i(1-t,x)$ *given by the* generalized Nelson equation

$$\boxed{\widehat{b}_i(t,x) + b_i(t,x) \;=\; \sum_{j=1}^{n} \Big(D_j\, \mathrm{a}_{ij}(t,x) + \mathrm{a}_{ij}(t,x) \cdot D_j \log \mathfrak{p}_t(x) \Big) \,, \quad i = 1, \cdots, n \,.} \tag{5.2}$$

Equivalently, we write

$$\boxed{\widehat{b}(t,x) + b(t,x) \;=\; \mathrm{div}\big(\mathrm{a}(t,x)\big) + \mathrm{a}(t,x) \cdot D \log \mathfrak{p}_t(x) \,,} \tag{5.3}$$

where $\mathrm{div}(\mathrm{a}(t,x)) := \big(\sum_{j=1}^{n} D_j\, \mathrm{a}_{ij}(t,x) \big)_{1 \le i \le n}$.

As Millet et al. (1989) show, this result can be extended to the case where we only assume the sum of the distributional derivatives $\sum_{j=1}^{n} D_j \big(\mathrm{a}_{ij}(t,x)\, \mathfrak{p}(t,x) \big)$, $i = 1, \cdots, n$ are locally integrable functions of $x \in \mathbb{R}^n$; see Example 8.1 below, as well as Russo et al. (2001).

5.1 Two diffuse initial distributions

Let us assume that we have started the system (2.1) with some given probability distribution P that has density $\mathfrak{p}_0(\cdot)$, for the initial random variable $\boldsymbol{\xi}$. This probability density function evolves then according to the forward Kolmogorov equation (2.3), giving rise to an entire family, or "flow", $\big(\mathfrak{p}_t(\cdot)\big)_{t \ge 0}$ of probability density functions.

Now suppose that we do this whole thing all over again, this time with a new distribution Q for the initial random variable $\boldsymbol{\xi}$ that has probability density function $\mathfrak{q}_0(\cdot)$. Then we get another temporal flow $\big(\mathfrak{q}_t(\cdot)\big)_{t \ge 0}$ of probability density functions, governed once again by (2.3). To avoid singularity issues, let us assume that both $\mathfrak{p}_0(\cdot)$ and $\mathfrak{q}_0(\cdot)$ are strictly positive on the entire real line.

Back in (5.2) and (5.3), we have now two families of vector fields, $\widehat{b}^{P}(t,\cdot)$ and $\widehat{b}^{Q}(t,\cdot)$ for $0 \le t < \infty$, corresponding to each of those initial probability distributions. It is clear from (5.3), that these vector fields are related via

$$\widehat{b}^{P}(t,x) - \widehat{b}^{Q}(t,x) \;=\; \mathrm{a}(t,x) \cdot D \log \left(\frac{\mathfrak{p}_t(x)}{\mathfrak{q}_t(x)} \right) \,, \qquad 0 \le t < \infty \,, \;\; x \in \mathbb{R}^n \,.$$

Likewise in (5.1), we get a backwards Brownian motion $B^P(\cdot)$ in the manner of (4.3) corresponding to the initial distribution P, and another Brownian motion $B^Q(\cdot)$ corresponding to the initial distribution Q. It should be clear from (4.3) or (5.1) how these are related; for instance, (5.1) gives

$$\mathrm{s}\big(1-\theta, X(1-\theta)\big)\left[\,\mathrm{d}B^Q(t)-\mathrm{d}B^P(t)\,\right] \;=\; \Big(\widehat{b}^P-\widehat{b}^Q\Big)\big(1-\theta,\widehat{X}(\theta)\big)\,\mathrm{d}\theta$$

$$=\; \mathrm{a}\big(1-\theta, X(1-\theta)\big)\;\cdot D\log\left(\frac{\mathfrak{p}_{1-\theta}}{\mathfrak{q}_{1-\theta}}\big(X(1-\theta)\big)\right).$$

In particular, in the case of invertible $\mathrm{s}(\cdot,\cdot)$, we have

$$B^Q \;=\; B^P+\int_0^{\cdot}\mathrm{s}'\big(1-\theta, X(1-\theta)\big)\;\cdot D\log\left(\frac{\mathfrak{p}_{1-\theta}}{\mathfrak{q}_{1-\theta}}\big(X(1-\theta)\big)\right)\mathrm{d}\theta\,.$$

6 Examples

The easiest example is that of standard Brownian motion on the real line: suppose $n=1$, $b\equiv 0$, $\mathrm{s}\equiv 1$ and $\xi\in\mathbb{R}$ is a real (non-random) constant. Then

$$\mathfrak{p}_t(x) \;=\; \big(2\pi t\big)^{-1/2}\exp\left\{-\frac{(x-\xi)^2}{2\,t}\right\},\qquad \frac{\partial}{\partial x}\log\mathfrak{p}_t(x) \;=\; \frac{\xi-x}{t} \;=:\; \widehat{b}(t,x)\,;$$

thus (5.1) becomes

$$\widehat{X}(t) \;=\; \widehat{X}(0)+\int_0^t\frac{\widehat{X}(1)-\widehat{X}(s)}{1-s}\,\mathrm{d}s\,+\,B(t)\,,\qquad 0\le t\le 1\,, \tag{6.1}$$

where $B(\cdot)$ is standard Brownian motion and, of course, $\widehat{X}(1)=\xi$. This is the equation for the Brownian Bridge. Note that the drift in this equation satisfies

$$\mathbb{E}\int_0^t\Big(\frac{\xi-\widehat{X}(s)}{1-s}\Big)^2\mathrm{d}s\;<\;\infty\qquad\text{for }\;0\le t<1\,,\;\;\text{but not for }\;t=1\,.$$

Indeed, with $\xi=0$ we have for $0\le t<1$:

$$\mathbb{E}\int_0^t\Big(\frac{\widehat{X}(s)}{1-s}\Big)^2\mathrm{d}s \;=\;\mathbb{E}\int_0^t\Big(\frac{X(1-s)}{1-s}\Big)^2\mathrm{d}s\;=\;\int_0^t\frac{\mathrm{d}s}{1-s}\;=\;\log\big(1/(1-t)\big)\;<\;\infty\,.$$

Example 6.1. Squared Bessel Process: Let us try something more elaborate. Fix $\delta>0$ and look at the diffusion

$$X(t)\;=\;\delta\,t+2\int_0^t\sqrt{X(s)}\,\mathrm{d}W(s)\,,\qquad 0\le t<\infty\,.$$

This is a diffusion with state space $[0,\infty)$, the square of a Bessel process in dimension δ (when δ is a natural number). We have of course $b(x)\equiv\delta$, $\mathrm{a}(x)=4x$ for $x\ge 0$ and

$$\mathfrak{p}_t(x)\;=\;\big(2t\big)^{-\delta/2}\big(\Gamma(\delta/2)\big)^{-1}x^{(\delta/2)-1}\,e^{-x/2t}\,,\qquad x\ge 0\,,\;\;t>0$$

(cf. Revuz & Yor (1999), page 441). Direct computation shows now that the time-reversed process $\widehat{X}(\cdot)$ of (1.1) has the semimartingale representation

$$\widehat{X}(t) \,=\, \widehat{X}(0) + \delta\, t - 2 \int_0^t \frac{\widehat{X}(s)}{1-s}\, \mathrm{d}s \,+\, 2 \int_0^t \sqrt{\widehat{X}(s)}\, \mathrm{d}B(s)\,, \qquad 0 \le t \le 1$$

of a "squared Bessel bridge" (cf. Revuz & Yor (1999), Exercise 3.11 on page 468).

Example 6.2. The Schmoluchowski Equation: Suppose $\mathrm{s} \equiv I_n$ is the unit $(n \times n)$ matrix, and that $b(t,x) \equiv D\,\Phi(x)$ is the gradient of a function $\Phi : \mathbb{R}^n \to \mathbb{R}$, the so-called *potential,* which is continuous, twice continuously differentiable, and satisfies

$$Z \,:=\, \int_{\mathbb{R}^n} e^{\,2\Phi(x)}\, dx < \infty\,.$$

Then the stochastic equation

$$X(t) \,=\, X(0) + \int_0^t D\,\Phi\,(X(s))\, \mathrm{d}s \,+\, W(t)\,, \qquad 0 \le t \le 1 \tag{6.2}$$

of (2.1) admits a pathwise unique, strong solution, which has a unique invariant probability measure given as

$$\mu(\mathrm{d}x) \,=\, \big(1/Z\big)\, e^{\,2\Phi(x)}\, \mathrm{d}x \qquad \text{on} \;\; \mathcal{B}(\mathbb{R}^n)\,; \tag{6.3}$$

see Karatzas & Shreve (1991), Exercise 5.6.18. If we assign this distribution to the initial random variable $\boldsymbol{\xi} = X(0)$, the random variable $X(t)$ has this same probability density function

$$\mathfrak{q}_t(x) \,\equiv\, \mathfrak{q}(x) \,=\, \big(1/Z\big)\, e^{\,2\Phi(x)}\,, \qquad x \in \mathbb{R}^n \tag{6.4}$$

at all times $t \in [0,1]$. It is not hard to see, that this function solves the stationary version (2.4) of the forward Kolmogorov equation.

It follows then from (5.1), (5.2) that the backward drift function $\widehat{b}(t,x) \equiv D\,\Phi(x)$ is exactly the same as the forward, and the time-reversed diffusion

$$\widehat{X}(t) \,=\, \widehat{X}(0) + \int_0^t D\,\Phi\,(\widehat{X}(s))\, \mathrm{d}s \,+\, B(t)\,, \qquad 0 \le t \le 1\,,$$

has the exact same drift and dispersion characteristics as the forward one. Here $B(\cdot)$ is a Brownian motion with respect to the backward filtration, and is independent of the random variable $\widehat{X}(0 = X(1)$.

We also note from the Corollary to Theorem 4.1 that

$$V(t) \,=\, W(t) + 2 \int_0^t D\,\Phi\,(X(s))\, \mathrm{d}s \,=\, X(t) - X(0) + \int_0^t D\,\Phi\,(X(s))\, \mathrm{d}s\,, \qquad 0 \le t \le 1$$

is a standard Brownian motion in its own filtration, and is independent of the random variable $X(1)$, though not of the random variable $X(0)$. Comparing this equation with (4.8) we deduce the rather remarkable fact that, for a Schmoluchowski-type diffusion process $X(\cdot)$ with invariant probability density function as in (6.4), *both* processes

$$X(t) - X(0) \pm \int_0^t D\Phi\big(X(s)\big)\,\mathrm{d}s\,, \qquad 0 \le t \le 1$$

are standard $n-$dimensional Brownian motions, each with respect to its own filtration. And they have the following, rather amusing, property: the one corresponding to the $(-)$ sign is independent of the initial condition $X(0)$, whereas the one corresponding to the $(+)$ sign is independent of the terminal condition $X(1)$.

As shown in Proposition 4 of Chatterjee & Pal (2010), this analysis extends to functions $\Phi : \mathbb{R}^n \to \mathbb{R}$ which are simply continuous and satisfy $\int_{\mathbb{R}^n} e^{2\Phi(x)}\,dx < \infty$, and to drift functions $b : \mathbb{R}^n \to \mathbb{R}$ for which $b(\cdot) = D\Phi(\cdot)$ holds in the sense of distributions.

Example 6.3. The Ornstein-Uhlenbeck Process: Suppose the equation of (2.1) is of the form

$$X(t) \;=\; X(0) - \alpha \int_0^t X(s)\,\mathrm{d}s \,+\, \sigma\, W(t)\,, \qquad 0 \le t \le 1$$

with $n = 1$ and $\alpha\,,\,\sigma$ positive constants. This linear equation admits a pathwise unique, strong solution, whose invariant distribution is the zero-mean Gaussian

$$\mathfrak{p}(x) \;=\; \big(2\pi v\big)^{-1/2} \exp\left\{-\frac{x^2}{2\,v}\right\} \qquad \text{with variance } \; v = \sigma^2/(2\alpha)\,.$$

If we assign this distribution to the initial random variable $\xi = X(0)$, and recall that $b(t,x) = -\alpha\, x$ is the forward drift, the equation (5.2) gives the expression $\widehat{b}(t,x) = -\alpha\, x$ for the backwards drift.

Once again, the process looks exactly the same forwards in time and backwards. In the context of Example 6.2 with $\sigma = 1$, this corresponds to a potential $\Phi(x) = -\,\alpha\, x^2/2\,,\;\; x \in \mathbb{R}$; in this case, let us note from the Corollary to Theorem 4.1 that the process

$$V(t) \;=\; X(t) - X(0) - \alpha \int_0^t X(s)\,\mathrm{d}s\,, \qquad 0 \le t \le 1$$

is standard Brownian motion with respect to its own filtration and independent of the random variable $X(1)$, though not of the random variable $X(0)$.

Example 6.4. Filtering an Unobserved Brownian Drift: Here is a generalization of Remark 4.2. Consider a standard Brownian motion process $W(\cdot)$ on the real line, and an independent random variable Θ with distribution μ that satisfies

$$\int_{\mathbb{R}} e^{\,y\vartheta} \mu(d\vartheta) < \infty \quad \text{ for all } \; y \in \mathbb{R}\,.$$ [1]

We take $\mathbb{F} = \{\mathcal{F}(t)\}_{0 \le t < \infty}$ with $\mathcal{F}(t) := \sigma(\Theta, X(s);\, 0 \le s \le t)$ as the underlying filtration.

[1] I am making this assumption only out of laziness; the entire development below should go through under far weaker conditions.

Suppose further that neither $W(\cdot)$ nor Θ are observable, but that the sum

$$X(t) \,=\, \Theta\, t \,+\, W(t)\,, \qquad 0 \le t < \infty$$

is. If we denote by $\mathbb{F}^X = \{\mathcal{F}^X(t)\}_{0 \le t < \infty}$ with $\mathcal{F}^X(t) := \sigma(X(s),\, 0 \le s \le t)$ the filtration generated by this process, then it can be shown (e.g., Beneš et al. (1991)) by the methods of linear filtering that the process $X(\cdot)$ is an $\mathbb{F}^X-$diffusion of the form

$$X(t) \,=\, \int_0^t G(u, X(u))\, \mathrm{d}u \,+\, N(t)\,, \qquad 0 \le t < \infty\,. \tag{6.5}$$

Here the "innovations process" $N(\cdot)$ is a standard $\mathbb{F}^X-$Brownian motion, and for $(t,x) \in (0,\infty) \times \mathbb{R}$ we have defined

$$G(t,x) \,:=\, \frac{\partial}{\partial x} \log F(t,x)\,, \qquad F(t,x) \,:=\, \int_{\mathbb{R}} e^{\,y\,x - (t/2)\,y^2}\, \mu(\mathrm{d}y)\,. \tag{6.6}$$

The equation (6.5) will play the role of the equation (2.1) in the context of the present example.

Now, the stochastic equation (6.5) – with $N(\cdot)$ as putative input, and $X(\cdot)$ as putative output – has a strong solution, so the "innovations process" $N(\cdot)$ and the the "observations process" $X(\cdot)$ generate the same filtration. We also have the filtering interpretation

$$\widehat{\Theta}(t) \,:=\, \mathbb{E}\big(\Theta \,\big|\, \mathcal{F}^X(t)\big) \,=\, G\big(t, X(t)\big)\,, \qquad 0 \le t < \infty\,.$$

It can be shown that the transition probabilities for this diffusion are given for $(x,\xi) \in \mathbb{R}^2$ as

$$\mathbb{P}\big(X(t) \in \mathrm{d}x \,\big|\, X(s) = \xi\big) \,=\, \frac{F(t,x)}{F(s,\xi)} \cdot \frac{\mathrm{d}x}{\sqrt{2\pi(t-s)}}\, \exp\left\{ -\frac{(x-\xi)^2}{2(t-s)} \right\}, \qquad s \le t < \infty\,.$$

Let us fix now $s = 0$ and $\xi \in \mathbb{R}$, and consider the diffusion process

$$X(t) \,=\, \xi + \int_0^t G\big(u, X(u)\big)\, \mathrm{d}u \,+\, N(t)\,, \qquad 0 \le t \le 1\,. \tag{6.7}$$

Then the probability density function of the random variable $X(t)$ is given as

$$\mathfrak{p}(t,x) \,=\, \frac{F(t,x)}{F(0,\xi)} \cdot \frac{1}{\sqrt{2\pi t}}\, \exp\left\{ -\frac{(x-\xi)^2}{2t} \right\}, \qquad 0 < t \le 1\,,$$

and thus

$$\frac{\partial}{\partial x}\, \log \mathfrak{p}(t,x) \,=\, G(t,x) - \frac{x-\xi}{t}\,.$$

From the equation (5.2) and $b(t,x) \equiv G(t,x)$ we deduce $\widehat{b}(t,x) = (\xi - x)/t$, so the time-reversal $\widehat{X}(\cdot) \equiv X(1-\cdot)$ of the process in (6.7) is of the form (6.1), namely, a Brownian Bridge:

$$\widehat{X}(t) \,=\, \widehat{X}(0) + \int_0^t \frac{\xi - \widehat{X}(s)}{1-s}\, \mathrm{d}s \,+\, B(t)\,, \qquad 0 \le t \le 1\,.$$

Remark 6.1. Let us consider again the transition probability density function

$$\mathcal{P}(s,\xi;t,x) \;=\; \frac{F(t,x)}{F(s,\xi)} \cdot \frac{1}{\sqrt{2\pi(t-s)}} \,\exp\left\{ -\frac{(x-\xi)^2}{2(t-s)} \right\}, \qquad s<t\le 1\,, \quad (x,\xi)\in\mathbb{R}^2$$

as above, and observe

$$\frac{\partial}{\partial\xi}\,\log\,\mathcal{P}(s,\xi;t,x) \;=\; -G(s,\xi)+\frac{x-\xi}{t-s}\,.$$

From the theory of Björk & Johannson (1992, 1996), we know that

$$Q(t) \;:=\; \frac{\partial}{\partial\xi}\,\log\,\mathcal{P}\big(0,\xi\,;\,t,X(t)\big) \;=\; \frac{X(t)-\xi}{t}\,, \qquad 0<t\le 1$$

is then a *reverse* martingale for the process of (6.7). This is reflected in the fact that, for the time-reversed process $\widehat{X}(\cdot)$ of (6.1), we have the $\widehat{\mathbb{F}}-$martingale dynamics

$$\mathrm{d}\left(\frac{\widehat{X}(t)-\xi}{1-t}\right) \;=\; \big(\widehat{X}(t)-\xi\big)\,\mathrm{d}\left(\frac{1}{1-t}\right)+\frac{1}{1-t}\,\mathrm{d}\,\widehat{X}(t) \;=\; \frac{\mathrm{d}B(t)}{1-t}\,, \qquad 0\le t<1\,.$$

Remark 6.2. With $\xi=0$, let us note from the Corollary to Theorem 4.1 that the process

$$\begin{aligned} V(t) \;&:=\; N(t)+\int_0^t\left(G\big(s,X(s)\big)-\frac{X(s)}{s}\right)\mathrm{d}s \;=\; X(t)-\int_0^t\frac{X(s)}{s}\,\mathrm{d}s \\ &=\; W(t)-\int_0^t\frac{W(s)}{s}\,\mathrm{d}s\,, \qquad 0\le t\le 1 \end{aligned} \tag{6.8}$$

is standard Brownian motion with respect to its own filtration, independent of the random variable $X(1)=W(1)+\Theta$. In fact, for every $t\in[0,1]$, the $\sigma-$algebra $\mathcal{F}^V(t)$ is independent of the $\sigma-$algebra $\sigma(X(u),\; t\le u\le 1)$. Thus, the process of (6.8) is a *generalization of the Jeulin-Yor process* in the present context.

Furthermore, quite unlike the stochastic equation (6.5) which admits a pathwise unique, strong solution, the equation

$$X(t) \;=\; \int_0^t\frac{X(s)}{s}\,\mathrm{d}s+V(t)\,, \qquad 0\le t\le 1$$

admits no strong solution. It can be considered as a "kinder, gentler, memoryless" version of the Tsirel'son (1975) equation, with instantaneous but unbounded drift function, which admits a unique-in-distribution weak solution but no strong solution. [2]

[2] I cannot help here but recall the great excitement that the appearance of the Tsirel'son example had generated in the mid-to-late 70's. Back then, one big problem in stochastic analysis was the so-called "innovations conjecture" – the question whether the innovations and observations processes carried always the same information (i.e., generated the same filtration), or not. All examples known at the time (by 1975) indicated that they did; Tsirel'son's example was the first indication that they might not.

Of course, as was being pointed out repeatedly in the heated debates of that era, "there is no filtering going on in Tsirel'son's example". In the neat special case discussed in Remark 6.3 of this example here, there *is* actual filtering going on, namely, of a symmetric Gaussian random variable observed in a bath of white noise. As the variance of this Gaussian signal goes to infinity, the innovations processes converge to the Jeulin-Yor process, and the strength of the stochastic equation (6.5) is lost in the infinite-variance limit.

Remark 6.3. *Symmetric Gaussian:* In the special case where Θ has a Gaussian distribution with mean zero and variance $k^2 \in (0, \infty)$. We write schematically $\Theta = k\,Z$ with Z standard Gaussian, and $X_k(t) = k\,\Theta\,t + W(t)$. Then, the function of (6.6) becomes

$$G_k(t, x) \;=\; \frac{x}{(1/k^2) + t}\,.$$

Whereas, the innovations Brownian motion process from (6.5) is, in accordance with Remark 4.2:

$$N_k(t) \;=\; X_k(t) - \int_0^t \frac{X_k(s)}{(1/k^2) + s}\,\mathrm{d}s \;=\; W(t) + k\,Z\,t - \int_0^t \frac{W(s) + k\,Z\,s}{(1/k^2) + s}\,\mathrm{d}s$$

$$= W(t) + \frac{Z}{k}\,\log\left(1 + k^2\,t\right) - \int_0^t \frac{W(s)}{(1/k^2) + s}\,\mathrm{d}s\,, \qquad 0 \le t \le 1$$

Letting $k \uparrow \infty$ in this expression we recover the Jeulin-Yor process of (4.12), (6.8), namely,

$$V(t) \;=\; W(t) - \int_0^t \frac{W(s)}{s}\,\mathrm{d}s\;, \qquad 0 \le t \le 1\,.$$

We see in other words, that this latter process $V(\cdot)$ plays the rôle of the innovations in a degenerate situation, where a signal of infinite variance is observed in a bath of additive white noise. And contrary to the situation encountered in Example 4.1 (which corresponds formally to $k = \infty$ here), the stochastic equation

$$X_k(t) \;=\; \int_0^t \frac{X_k(s)}{(1/k^2) + s}\,\mathrm{d}s \;+\; N_k(t)\,, \qquad 0 \le t \le 1$$

does have a pathwise unique, strong solution: the innovations process $N_k(\cdot)$ and the observations process $X_k(\cdot)$ generate exactly the same filtration.

7 Strict Time-Reversibility

Let us agree to call a diffusion process $X(\cdot)$ as in (2.1) *strictly time-reversible,* if its time reversal $\widehat{X}(\cdot)$ of (1.1) is not just a diffusion process as in (5.1), but has actually the same local drift and dispersion characteristics as the original process $X(\cdot)$; in other words,

$$\widehat{b}_i(t, x) \;=\; b_i(t, x)\,, \qquad i = 1, \cdots, n\,, \quad (t, x) \in [0, 1] \times \mathbb{R}^n$$

for the functions $\widehat{b}_i(\cdot\,,\cdot)$ of Theorem 5.1. This is the situation encountered in the Examples of 6.2 and 6.3.

Now, in light of the generalized Nelson equation (5.2), this requirement amounts to

$$b_i(t, x) \;=\; \frac{1}{2}\sum_{j=1}^{n}\Big(D_j\,\mathrm{a}_{ij}(t, x) + \mathrm{a}_{ij}(t, x)\cdot D_j \log \mathfrak{p}_t(x)\Big)\,, \qquad i = 1, \cdots, n\,. \tag{7.1}$$

Substituting these expressions into the forward Kolmogorov equation (2.3) for the probability density function $\mathfrak{p}_t(\cdot) : \mathbb{R}^n \to (0,\infty)$, we obtain

$$\frac{\partial}{\partial t}\,\mathfrak{p}_t(x) = \frac{1}{2}\sum_{i=1}^n D_i\left(\sum_{j=1}^n \Big\{ D_j\big(\mathrm{a}_{ij}(t,x)\,\mathfrak{p}_t(x)\big) - \mathrm{a}_{ij}(t,x)\,D_j\,\mathfrak{p}_t(x) - \mathfrak{p}_t(x)\,D_j\,\mathrm{a}_{ij}(t,x)\Big\}\right) = 0\,.$$

In other words, the diffusion process $X(\cdot)$ has to possess an invariant probability measure; and $\mathfrak{p}_t(\cdot) \equiv \mathfrak{p}(\cdot)$ has to be the density of this measure and to satisfy equation (2.4).

7.1 Time-Homogeneity

This suggests considering drift $b_i(\cdot)$ and covariation $\mathrm{a}_{ij}(\cdot)$ functions which are time-homogeneous and satisfy

$$b(\cdot) \equiv \beta(\cdot)\,, \quad \text{where} \quad \beta_i(x) := \sum_{j=1}^n \frac{D_j\big(\mathrm{a}_{ij}(x)\mathfrak{p}(x)\big)}{2\,\mathfrak{p}(x)} = \frac{1}{2}\sum_{j=1}^n \Big(D_j\,\mathrm{a}_{ij}(x) + \mathrm{a}_{ij}(x)\cdot D_j \log \mathfrak{p}(x)\Big) \tag{7.2}$$

for $i = 1, \cdots, n$, or equivalently

$$\boxed{\beta(x) := \frac{1}{2}\Big(\operatorname{div}\mathrm{a}(x) + \mathrm{a}(x)\,D\log\mathfrak{p}(x)\Big)\,.} \tag{7.3}$$

Theorem 7.1. *Suppose we are given a smooth function $\mathrm{a} : \mathbb{R}^n \to \mathbb{S}^n$ and a smooth probability density function $\mathfrak{p} : \mathbb{R}^n \to (0,\infty)$. The diffusion process with covariation matrix function $\mathrm{a}(\cdot)$ and drift vector function $\beta(\cdot)$ as in (7.3), is the unique strictly time-reversible diffusion process with covariation matrix function $\mathrm{a}(\cdot)$ and invariant measure with probability density function $\mathfrak{p}(\cdot)$.*

Proof: In view of the preceding discussion, all that remains is to check the invariance of $\mathfrak{p}(\cdot)$ for this diffusion, that is, the stationary version (2.4) of the forward Kolmogorov equation. But now (7.2) gives

$$2\,D_i\big(\beta_i(x)\,\mathfrak{p}(x)\big) = D_i \sum_{j=1}^n \mathfrak{p}(x)\Big(D_j\,\mathrm{a}_{ij}(x) + \mathrm{a}_{ij}(x)\cdot D_j\log\mathfrak{p}(x)\Big)$$

$$= \sum_{j=1}^n \Big(D_i\,\mathfrak{p}(x)\,D_j\,\mathrm{a}_{ij}(x) + \mathfrak{p}(x)\,D^2_{ij}\,\mathrm{a}_{ij}(x) + D_j\,\mathfrak{p}(x)\,D_i\,\mathrm{a}_{ij}(x) + \mathrm{a}_{ij}(x)\,D^2_{ij}\,\mathfrak{p}(x)\Big)\,,$$

and adding up over $i = 1, \cdots, n$ we obtain the forward Kolmogorov equation (2.4), namely

$$2\sum_{i=1}^n D_i\big(\beta_i(x)\,\mathfrak{p}(x)\big) = \sum_{i=1}^n\sum_{j=1}^n D^2_{ij}\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\big)\,. \qquad \square$$

7.2 Forward and Backward Motions in Duality

Let us suppose that the diffusion process of (2.1) possesses an invariant measure with density function $\mathfrak{p}(\cdot)$, and look at the generators

$$(\mathcal{L}f)(x) \;:=\; \frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{a}_{ij}(x)\,D^2_{ij}f(x)\;+\;\sum_{i=1}^{n} b_i(x)\,D_i f(x) \tag{7.4}$$

and

$$\big(\widehat{\mathcal{L}}f\big)(x) \;:=\; \frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{a}_{ij}(x)\,D^2_{ij}f(x)\;+\;\sum_{i=1}^{n} \widehat{b}_i(x)\,D_i f(x) \tag{7.5}$$

of the forward and backward diffusions, respectively, in the time-homogeneous case.

We claim that these generators are formal adjoints of each other, with respect to the invariant measure: for any two $\mathcal{C}^2(\mathbb{R}^n)$ functions $f(\cdot)$ and $g(\cdot)$ with compact support, we have the duality relationship

$$\int_{\mathbb{R}^n} f(x)\,(\mathcal{L}g)(x)\,\mathfrak{p}(x)\,\mathrm{d}x \;=\; \int_{\mathbb{R}^n} g(x)\,\big(\widehat{\mathcal{L}}f\big)(x)\,\mathfrak{p}(x)\,\mathrm{d}x\,. \tag{7.6}$$

The proof is a straightforward computation: the left-hand side of (7.6) is equal to

$$\int_{\mathbb{R}^n} f(x)\,\mathfrak{p}(x)\left(\frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{a}_{ij}(x)\,D^2_{ij}\,g(x)\;+\;\sum_{i=1}^{n} b_i(x)\,D_i\,g(x)\right)\mathrm{d}x$$

$$= \int_{\mathbb{R}^n} g(x)\left(\frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n} D^2_{ij}\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\,f(x)\big)\;-\;\sum_{i=1}^{n} D_i\big(b_i(x)\,\mathfrak{p}(x)\,f(x)\big)\right)\mathrm{d}x$$

$$= \int_{\mathbb{R}^n} g(x)\,f(x)\bigg(\frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n} D^2_{ij}\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\big)\;-\;\sum_{i=1}^{n} D_i\big(b_i(x)\,\mathfrak{p}(x)\big)\bigg)\mathrm{d}x$$

$$+\int_{\mathbb{R}^n} g(x)\left(\frac{1}{2}\,\sum_{i=1}^{n}\sum_{j=1}^{n}\Big[2D_i f(x)\,D_j\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\big)\;+\;\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\cdot D^2_{ij}f(x)\Big]\right)\mathrm{d}x$$

$$-\int_{\mathbb{R}^n} g(x)\,\mathfrak{p}(x)\Big(\sum_{i=1}^{n} b_i(x)\,D_i f(x)\big)\Big)\mathrm{d}x$$

$$= \int_{\mathbb{R}^n} g(x)\,\mathfrak{p}(x)\left[\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\mathrm{a}_{ij}(x)}{2}D^2_{ij}f(x)+\sum_{i=1}^{n}\Big\{\frac{D_j\big(\mathrm{a}_{ij}(x)\,\mathfrak{p}(x)\big)}{\mathfrak{p}(x)}-b_i(x)\Big\}D_i f(x)\right]\mathrm{d}x$$

$$= \int_{\mathbb{R}^n} g(x)\,\mathfrak{p}(x)\left[\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\mathrm{a}_{ij}(x)}{2}D^2_{ij}f(x)+\sum_{i=1}^{n}\widehat{b}_i(x)D_i f(x)\right]dx = \int_{\mathbb{R}^n} g(x)\,\mathfrak{p}(x)\,\big(\widehat{\mathcal{L}}f\big)(x)\,\mathrm{d}x\,.$$

We have used integration by parts, the forward Kolmogorov equation (2.4), as well as the equations (7.2), (7.5).

8 Covariances and Potentials

The considerations in Example 6.2 suggest that a large class of strictly time-reversible diffusion processes $X(\cdot)$ can be generated in the following manner: pick smooth functions $\Phi : \mathbb{R}^n \to \mathbb{R}$ and $\mathrm{s} : \mathbb{R}^n \to \mathbb{S}^n$ with

$$Q := \int_{\mathbb{R}^n} e^{2\Phi(\xi)} \, \mathrm{d}\xi < \infty \,, \qquad \text{and set} \quad \mathfrak{p}(x) := \big(1/Q\big)\, e^{2\Phi(x)} \,, \qquad x \in \mathbb{R}^n \tag{8.1}$$

as well as

$$\mathrm{a}(\cdot) := \mathrm{s}(\cdot)\,\mathrm{s}'(\cdot) \,, \qquad b_i(x) := \frac{1}{2} \sum_{j=1}^{n} D_j \, \mathrm{a}_{ij}(x) + \Big(\mathrm{a}(x) \cdot D\Phi(x)\Big)_i \,, \quad i = 1, \cdots, n \,, \tag{8.2}$$

the latter in accordance with (7.2). Then the diffusion process $X(\cdot)$ of (2.1) is strictly time-reversible, and has invariant measure with probability density function given by $\mathfrak{p}(\cdot)$ of (8.1). Let us also note that the situation of Example 6.2 is a special case of this more general structure.

An interesting structure emerges from (8.2): it mandates that the so-called *Fichera drift* vector field $\mathfrak{b}(\cdot) := b(\cdot) - (1/2)\,\mathrm{div}(\mathrm{a}(\cdot))$ given by

$$\mathfrak{b}_i(x) := b_i(x) - \frac{1}{2} \sum_{j=1}^{n} D_j \, \mathrm{a}_{ij}(x) \,, \qquad i = 1, \cdots, n$$

(Friedman (1975), p. 208), be the inner product of the covariance matrix with a vector field which is a gradient in the range of the covariance matrix function: $\mathfrak{b}(x) = \mathrm{a}(\cdot)\, D\Phi(\cdot)$.

I find the appearance of the Fichera drifts in this context quite intriguing. They play important rôles in deciding the attainability of lower-dimensional manifolds by diffusion processes (Friedman (1975)), as well as the existence or absence of arbitrage (Fernholz & Karatzas (2010)). Here, they arise in the context of time reversibility. Perhaps I should not find this surprising, given the understanding of the "adjoint Markov process" in terms of time reversal due to Nelson (1958).

8.1 The One-Dimensional Case

In dimension $n = 1$ the second expression of (8.2) becomes

$$2\,\Phi'(x) = \frac{2\,b(x) - \mathrm{a}'(x)}{\mathrm{a}(x)} = \frac{2\,b(x)}{\mathrm{a}(x)} + \left(\log\Big(\frac{1}{\mathrm{a}(x)}\Big)\right)'$$

and leads to a probability density function for the invariant measure of the form

$$\mathfrak{p}(x) = \frac{1}{Q}\, \exp\Big\{ \int_{\cdot}^{x} \frac{2\,b(y) - \mathrm{a}'(y)}{\mathrm{a}(y)}\, dy \Big\} = \frac{1}{Q^*}\, \frac{1}{\mathrm{a}(x)}\, \exp\Big\{ \int_{\cdot}^{x} \frac{2\,b(y)}{\mathrm{a}(y)}\, dy \Big\}$$

in (8.1). The second of these two expressions for $\mathfrak{p}(\cdot)$ agrees with the formula (5.5.51) for the density of the speed measure and with Exercise 5.5.40 in Karatzas & Shreve (1991); whereas the first of these two expressions is *almost* the one that appears on page 108 of Fukushima & Stroock (1986). It seems to me that they made a slight error as far as this formula is concerned, though I have to admit that I did not try to verify their formula (0.3), on the symmetry of the "infinitesimal generator", for this choice of density – or to refute it for theirs.

Remark 8.1. It is of interest to note the adjoint relationship

$$\mathfrak{p}(x)\, P_t(x,y) \;=\; \mathfrak{p}(y)\, P_t(y,x)\,; \qquad t\in(0,\infty) \;\; (x,y)\in\mathbb{R}^2$$

between the density of the invariant measure and the transition probability density function $P_t(\cdot\,,\cdot) \equiv \mathcal{P}(0,\cdot\,;\,t\,,\cdot\,)$ of the diffusion process, to wit: $\mathbb{P}(X(t)\,\in\,A\,|\,X(0)\,=\,\xi)\,=\,\int_A P_t(\xi\,,\eta)\,d\eta\,,$ $A\in\mathcal{B}(\mathbb{R})$. This adjoint relationship appears in Kolmogorov (1937) and Itô & McKean (1974); see also the recent work by Ekström & Tysk (2010).

Example 8.1. Bang-Bang Diffusion: Consider the function $\Phi(x) = -\lambda\,|x|/\sigma^2\,,\; x\in\mathbb{R}$ for some real constant $\lambda>0\,,\;\sigma>0\,$. Then the equation (8.2) takes the form

$$X(t) \;=\; X(0) - \lambda \int_0^t \mathrm{sgn}(X(s))\,\mathrm{d}s \;+\; \sigma\, W(t)\,, \qquad 0\le t\le 1$$

with (discontinuous, but locally integrable) "bang-bang" drift, and admits a pathwise unique, strong solution with double exponential invariant density

$$\mathfrak{p}(x) \;=\; \big(1/Q\big)\, e^{\,2\Phi(x)} \;=\; \frac{\varrho}{2}\, e^{\,-\varrho\,|x|}\,, \qquad x\in\mathbb{R} \qquad \text{with} \qquad \varrho \;:=\; 2\lambda/\sigma^2\,.$$

The transition probabilities of this diffusion can be computed explicitly; see Karatzas & Shreve (1984), or pages 437-441 in Karatzas & Shreve (1991). Started with its invariant distribution, the bang-bang diffusion $X(\cdot)$ is strictly time-reversible; so is its absolute value $|X(\cdot)|$, a reflected Brownian motion with local variance σ^2, negative local drift $-\lambda$, and invariant measure with exponential density $\varrho\, e^{-\varrho\,\xi}\,,\;\; \xi\in\mathbb{R}_+$ (cf. Theorem 2 in Graversen & Shiryaev (2000)).

On the other hand, as remarked earlier and with $\sigma=1$ for simplicity, both processes

$$X(t) - X(0) \pm \lambda \int_0^t \mathrm{sgn}\big(X(s)\big)\mathrm{d}s\,, \qquad 0\le t\le 1$$

are standard Brownian Motions, each in its own filtration. The one corresponding to the $(+)$ sign is independent of the initial condition $X(0)$, whereas the one corresponding to the $(-)$ sign is independent of the terminal condition $X(1)$.

9 Appendix: The Backward Itô Integral

For two continuous semimartingales $X(\cdot) = X(0){+}M(\cdot){+}B(\cdot)$ and $Y(\cdot) = Y(0){+}N(\cdot){+}C(\cdot)\,$, with $B(\cdot)\,, C(\cdot)$ continuous adapted processes of finite variation and $M(\cdot)\,, N(\cdot)$ continuous local martingales, let us recall the Definition 3.3.13 of the Fisk-Stratonovich integral from on page 156 of Karatzas & Shreve (1991), as well as its properties in Problems 3.3.14 and 3.3.15.

By analogy with this definition, we introduce the *Backwards Itô Integral*

$$\int_0^{\cdot} Y(t)\bullet \mathrm{d}X(t) \;:=\; \int_0^{\cdot} Y(t)\,\mathrm{d}M(t) + \int_0^{\cdot} Y(t)\,\mathrm{d}B(t) + \langle M,N\rangle(\cdot)\,,$$

where the first (respectively, the second) integral on the right-hand side is to be interpreted in the Itô (respectively, the Lebesgue-Stieltjes) sense.

If $\Pi = \{t_0, t_1, \cdots, t_m\}$ is a partition of the interval $[0, T]$ with $0 = t_0 < t_1 < \cdots < t_{m-1} < t_m = T$, then the sums

$$\sum_{j=0}^{m-1} Y(t_{j+1}) \big(X(t_{j+1}) - X(t_j) \big)$$

converge in probability to $\int_0^T Y(t) \bullet dX(t)$ as the mesh $\|\Pi\|$ of the partition tends to zero. Note that the increments of $X(\cdot)$ here "stick backwards into the past", as opposed to "sticking forward into the future" as in the Itô integral.

For this integral we have the change of variable formula

$$f(X(\cdot)) \;=\; f(X(0)) + \sum_{i=1}^n \int_0^\cdot D_i f(X(t)) \bullet \mathrm{d}X_i(t) - (1/2) \sum_{i=1}^n \sum_{j=1}^n \int_0^\cdot D^2_{ij} f(X(t)) \, \mathrm{d}\langle M_i, M_j \rangle(t) \,,$$

where now $X(\cdot) = (X_1(\cdot), \cdots, X_n(\cdot))'$ is a vector of continuous semimartingales of the form $X_i(\cdot) = X_i(0) + M_i(\cdot) + B_i(\cdot)$, $i = 1, \cdots, n$ as above. Note the change of sign, from (+) to (−) in the last, stochastic correction term.

Bibliography

ANDERSON, B.D.O. (1982) Reverse time diffusion equations. *Stochastic Processes & Their Applications* **12**, 313-326.

BANNER, A.D., FERNHOLZ, E.R. & KARATZAS, I. [BFK] (2005) Atlas models of equity markets. *Annals of Applied Probability* **15**, 2296-2330.

BASS, R. & PARDOUX, E. (1987) Uniqueness of diffusions with piecewise constant coëfficients. *Probability Theory and Related Fields* **76**, 557-572.

BELL, D. (1995) *Degenerate Stochastic Differential Equations & Hypoëllipticity.* Longman Group Ltd., Essex.

BENEŠ, V.E., KARATZAS, I. & RISHEL, R.W. (1991) The separation principle for a Bayesian adaptive control problem with no strict-sense optimal law. *Stochastics Monographs* **5**, 121-156.

BJÖRK, T. & JOHANSSON, B. (1992) Adaptive prediction and reverse martingales. *Stochastic Processes & Their Applications* **43**, 191-222.

BJÖRK, T. & JOHANSSON, B. (1996) Parameter estimation and reverse martingales. *Stochastic Processes & Their Applications* **63**, 235-263.

CARLEN, E.A. (1984) Conservative diffusions. *Comm. Math. Phys.* **94**, 293-315.

CASTANON, D.A. (1982) Reverse time diffusion processes. *IEEE Transactions on Information Theory* **IT-L8**, 953-956.

CHATTERJEE, S. AND PAL, S. (2010) A phase transition behavior for Brownian motions interacting through their ranks. *Probability Theory and Related Fields*, to appear.

CHATTERJEE, S. AND PAL, S. (2011) A combinatorial analysis of interacting diffusions. *Journal of Theoretical Probability,* to appear.

DAVIS, M.H.A. (1977) *Linear Estimation & Stochastic Control.* Halsted Press, J. Wiley & Sons, New York.

EKSTRÖM, E. & TYSK, J. (2010) Forward is backward for time-homogeneous diffusions. *Preprint, Uppsala University.*

FERNHOLZ, E.R. (2002) *Stochastic Portfolio Theory.* Springer Verlag, New York.

FERNHOLZ, D. & KARATZAS, I. (2010) On optimal arbitrage. *Annals of Applied Probability* **20**, 1179-1204.

FÖLLMER, H. (1985) An entropy approach to the time reversal of difusion processes. *Lecture Notes in Control and Information Systems* **69**, 156-163. Springer-Verlag, NY.

FÖLLMER, H. (1986) Time reversal in Wiener space. *Lecture Notes in Mathematics* **1158**, 119-129. Springer Verlag, New York.

FRIEDMAN, A. (1975) *Stochastic Differential Equations & Applications, Vol. 1.* Academic Press, New York.

FUKUSHIMA, M. & STROOCK, D.W. (1986) Reversibility of solutions to martingale problems. In "Probability, Statistical Physics and Number Theory". *Advances in Mathematics; Supplemental Studies Vol.* **9**, 107-122.

GRAVERSEN, S.E. & SHIRYAEV (2000) An extension of P. Lévy's distributional properties to the case of Brownian motion with drift. *Bernoulli* **6**, 615-620.

HAUSSMANN, U. & PARDOUX, E. (1986) Time reversal of diffusions. *Annals of Probability* **14**, 1188-1205.

HWANG, C.R., HWANG-MA, S.Y.& SHEU, S.J. (2005) Accelerating diffusions. *Annals of Applied Probability* **15**, 1433-1444.

ICHIBA, T., PAPATHANAKOS, V., BANNER, A.D., FERNHOLZ, E.R. & KARATZAS, I. (2011) Hybrid Atlas Models. *Annals of Applied Probability,* to appear.

ITÔ, K. & MCKEAN, H.P. JR. (1974) *Diffusions Processes and Their Sample Paths.* Second Printing, Corrected. Springer-Verlag, New York.

JACOD, J. (1977) A General Representation Theorem for Martingales. *Proceedings of Symposia in Pure Mathematics* **9**, 1-27. American Mathematical Society, Providence, RI.

JEULIN, TH. & YOR, M. (1990) Filtration des ponts browniens et équations différentielles stochastiques linéaires. *Lecture Notes in Mathematics* **1426**, 227-265. Springer Verlag, NY.

KARATZAS, I. & SHREVE, S.E. (1984) Trivariate density of Brownian motion, its local and occupation times, with application to stochastic control. *Annals of Probability* **12**, 819-828.

KARATZAS, I. & SHREVE, S.E. (1991) *Brownian Motion and Stochastic Calculus.* Second Edition, Springer Verlag, New York.

KOLMOGOROV, A.N. (1931) Über die analytischen Methoden in der Wahscheinlichkeitsrechnung. *Mathematische Annalen* **104**, 415-458.

KOLMOGOROV, A.N. (1937) Zur Umkehrkeit der statistischen Naturgesetze. *Mathematische Annalen* **113**, 766-772.

MEYER, P.A. (1994) Sur une transformation du mouvement brownien due à Jeulin et Yor. *Lecture Notes in Mathematics* **1583**, 98-101. Springer-Verlag, NY.

MILLET, A., NUALART, D. & SANZ, M. (1989) Integration by parts and time reversal for diffusion processes. *Annals of Probability* **17**, 208-238.

NELSON, E. (1958) The adjoint Markoff process. *Duke Mathematical Journal* **25**, 671-690.

NELSON, E. (1967) *Dynamical Theories of Brownian Motion.* Princeton University Press, NJ.

NUALART, D. (1995) *The Malliavin Calculus and Related Topics.* Springer-Verlag, New York.

PAL, S. & PITMAN, J. (2008) One-dimensional Brownian particle systems with rank-dependent drifts. *Annals of Applied Probability* **18**, 2179-2207.

PARDOUX, E. (1986) Grossissement d'une filtration et retournement du temps. *Lecture Notes in Mathematics* **1204**, 1188-1205. Springer Verlag, New York.

REVUZ, D. & YOR, M. (1999) *Continuous Martingales and Brownian Motion.* Second Edition, Springer Verlag, New York.

ROGERS, L.C.G. (1985) Smooth transition densities for one-dimensional diffusions. *Bull. London Math. Society* **17**, 157-161.

ROGERS, L.C.G. & WILLIAMS, D. (1994) *Diffusions, Markov Processes and Martingales. Volume I: Foundations.* Second Edition, J. Wiley & Sons, New York.

ROGERS, L.C.G. & WILLIAMS, D. (1987) *Diffusions, Markov Processes and Martingales. Volume II: Itô Calculus.* J. Wiley & Sons, New York.

RUSSO, F., VALLOIS, P. & WOLF, J. (2001) A generalized class of Lyons-Zheng processes. *Bernoulli* **7**, 363-378.

SCHRÖDINGER, E. (1931) Über die Umkehrung der Naturgesetze. *Sitzungsberichte der Preussischen Akademie der Wissenschaften: Physikalisch-Mathematischen Klasse,* 144-153.

SCHRÖDINGER, E. (1932) Sur la théorie relativiste de l'électron et l'interprétation de la mécanique quantique. *Ann. Inst. H. Poincaré* **2**, 269-310.

TSIREL'SON, B.S. (1975) An example of a stochastic differential equation having no strong solution. *Theory of Probability & Its Applications* **20**, 427-430.

WALSH, J.B. (1982) A non-reversible semimartingale. *Lecture Notes in Mathematics* **921**, 212. Springer Verlag, NY.